

mixmogam 2.0: Technical Details of the Mixed-Model Association Engine

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1 Overview and notation

mixmogam fits the standard linear mixed model for genome-wide association mapping,

$$\mathbf{y} = \mathbf{X}\beta + u + e, \quad u \sim \mathcal{N}(0, \sigma_g^2 \mathbf{K}), \quad e \sim \mathcal{N}(0, \sigma_e^2 I_n), \quad (1)$$

where n is the number of samples, $\mathbf{X} \in \mathbb{R}^{n \times q}$ the covariate design (an intercept is included by default), and $\mathbf{K} \in \mathbb{R}^{n \times n}$ a kinship (relationship) matrix. We write $\delta = \sigma_e^2 / \sigma_g^2$ and $\mathbf{V} = \sigma_g^2 (\mathbf{K} + \delta I_n)$. The covariance of (1) is parameterized by the single ratio δ ; all variance-component estimation reduces to one-dimensional optimization of the REML or ML profile likelihood over δ .

The package implements three regimes:

1. **Exact** ($n \lesssim 8,000$): one dense eigendecomposition of $\mathbf{K}$ and of the restricted matrix $\mathbf{S}(\mathbf{K} + I)\mathbf{S}$ (Section 2); the association scan evaluates the exact EMMAX statistic in batched dense GEMMs (Section 3).
2. **Truncated spectrum** (large n): a randomized top- k eigenbasis of $\mathbf{K}$ with an exact identity-tail correction (Section 5).
3. **SLQ fitting** (large n): variance components by deflated stochastic Lanczos quadrature, requiring only matrix-vector products with $\mathbf{K}$ and no $O(n^3)$ factorization (Section 6).

Genotypes are stored sample-major as $n \times m$ matrices of `int8` dosages in $\{0, 1, 2\}$ with -1 encoding missing calls; SNP blocks are converted to `float32` with fused mean imputation immediately before use, so the float working set is bounded by the block size regardless of m .

2 Variance-component estimation: exact EMMA

2.1 Restricted eigenspace

Following Kang et al. (2008), the REML likelihood can be expressed through the spectral decomposition of the *restricted* matrix $\mathbf{M} = \mathbf{S}(\mathbf{K} + I_n)\mathbf{S}$ with $\mathbf{S} = I_n - \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ (the v1 construction; eigenpairs beyond the q annihilated directions, offset by one). mixmogam evaluates the identical likelihood from a *single* eigendecomposition of $\mathbf{K}$ via the matrix determinant lemma: with $V = \mathbf{K} + \delta I_n = \mathbf{U} \text{diag}(\lambda_i + \delta) \mathbf{U}^\top$ and $W = \mathbf{U}^\top \mathbf{X}$, $w = \mathbf{U}^\top \mathbf{y}$,

$$\text{RSS}(\delta) = w^\top D w - b^\top A^{-1} b, \quad A = W^\top D W, \quad b = W^\top D w, \quad D = \text{diag}\left(\frac{1}{\lambda_i + \delta}\right), \quad (2)$$

$$\ell_R(\delta) = \frac{1}{2} \left[p \left(\log \frac{p}{2\pi} - 1 - \log \text{RSS} \right) - \left(\sum_i \log(\lambda_i + \delta) + \log |A| - \log |\mathbf{X}^\top \mathbf{X}| \right) \right], \quad (3)$$

with the derivative in closed form through $A' = -W^\top D^2 W$, $b' = -W^\top D^2 w$. The two formulations agree to 5×10^{-13} on likelihoods in the test suite; the lemma form replaces the second $O(n^3)$ eigendecomposition with $O(nq^2)$ work.

Algorithm 1 EMMA exact fit (REML shown; ML analogous)

Require: $\mathbf{y}$, $\mathbf{X}$, $\mathbf{K}$, grid $(G, l_{\min}, l_{\max})$, tolerance τ

- 1: $\mathbf{Q} \leftarrow$ thin QR of $\mathbf{X}$; $\mathbf{S} \leftarrow I - \mathbf{Q}\mathbf{Q}^\top$
 - 2: eigendecompose $\mathbf{M} = \mathbf{S}(\mathbf{K} + I)\mathbf{S}$; keep $\{\lambda_i^R, \eta_i\}_{i \leq p}$
 - 3: vectorize $\ell_R(\delta_j)$, $\ell'_R(\delta_j)$ over the grid $j = 0..G$
 - 4: $i^* \leftarrow \arg \max_j \ell_R(\delta_j)$
 - 5: find i with $\ell'_R(\delta_i) > 0 > \ell'_R(\delta_{i+1})$ maximizing mean endpoint likelihood
 - 6: $\hat{\delta} \leftarrow \text{Brent}(\ell'_R, \delta_i, \delta_{i+1}, \tau)$; accept per boundary rules, else fall back to δ_{i^*}
 - 7: **return** $\hat{\delta}$, $\hat{\sigma}_g^2 = s_1(\hat{\delta})/p$, h_{pseudo}^2 , β via (5)
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2.2 Profile likelihood and grid

The REML profile log-likelihood as a function of δ is The ML variant replaces p by n and drops the $\log |A| - \log |\mathbf{X}^\top \mathbf{X}|$ terms (same quadratic form). The equivalence with the restricted-eigenspace form above (equation label kept for cross-reference) is asserted in the tests; v1's deliberate factor-of-two in the derivative function – harmless for root finding – is normalized. The grid $\delta_j = \exp(\text{linspace}(l_{\min}, l_{\max}, G+1))$ with $G = 100$ and $[l_{\min}, l_{\max}] = [-10, 10]$ is evaluated *vectorized over all grid points simultaneously* by broadcasting λ^R against δ , so the whole grid costs $O(pG)$ fused operations. The maximizing bracket is located by a sign change of $d\ell/d\delta$ and polished with Brent's method (v1 used Newton from the bracket midpoint; the root is identical and Brent cannot diverge), with v1's boundary-acceptance and grid-argmax fallback rules retained. Finally

$$\hat{\sigma}_g^2 = \frac{s_1(\hat{\delta})}{p}, \quad \hat{\sigma}_e^2 = \hat{\delta} \hat{\sigma}_g^2, \quad h_{\text{pseudo}}^2 = \frac{1}{1 + \hat{\delta}}. \quad (4)$$

2.3 GLS coefficients and gBLUP

With $\equiv \mathbf{V}^{-1/2}$ (applied implicitly, Section 5),

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{y} \quad \text{computed as} \quad ((\mathbf{X})^\top (\mathbf{X}))^{-1} (\mathbf{X})^\top (\mathbf{y}), \quad (5)$$

and the gBLUP of breeding values is $\hat{u} = \hat{\sigma}_g^2 \mathbf{K} \mathbf{V}^{-1} (\mathbf{y} - \mathbf{X} \hat{\beta})$, with $\mathbf{V}^{-1}$ applied through the eigenbasis.

3 The association scan: EMMA as batched GEMM

3.1 Single-SNP statistic

For a candidate SNP $g \in \mathbb{R}^n$ (missing calls mean-imputed), v1 regressed the transformed, covariate-residualized phenotype on the transformed, covariate-residualized SNP by an explicit least-squares solve *per SNP*. The identical statistic has a closed form. Define, at the fitted $\hat{\delta}$,

$$\mathbf{Q} \leftarrow \text{qr}(\mathbf{X}), \quad r \leftarrow (I - \mathbf{Q}\mathbf{Q}^\top) \mathbf{y}, \quad \tilde{g} \leftarrow (I - \mathbf{Q}\mathbf{Q}^\top) g, \quad \text{RSS}_0 \leftarrow \|r\|^2. \quad (6)$$

The one-degree-of-freedom F statistic is the rank-one update

$$\text{RSS}(g) = \text{RSS}_0 - \frac{(\tilde{g}^\top r)^2}{\tilde{g}^\top \tilde{g}}, \quad F(g) = \left(\frac{\text{RSS}_0}{\text{RSS}(g)} - 1 \right) (n - q - 1), \quad p(g) = F_{1, n-q-1}(F(g)), \quad (7)$$

Algorithm 2 Batched EMMAX scan

Require: fitted $\hat{\delta}$; covariates $\mathbf{X}$; phenotype $\mathbf{y}$; genotype store returning blocks $\mathbf{G} \in \mathbb{R}^{k \times n}$; block size k , dtype

- 1: $\mathbf{Q} \leftarrow \text{qr}(\mathbf{X})$; $r \leftarrow (I - \mathbf{Q}\mathbf{Q}^\top)\mathbf{y}$; $\text{RSS}_0 \leftarrow \|r\|^2$; $\nu \leftarrow n - q - 1$
- 2: **for all** blocks $\mathbf{G}$ from the store **do**
- 3: $\tilde{\mathbf{G}}^\top \leftarrow (I - \mathbf{Q}\mathbf{Q}^\top) \mathbf{G}^\top$ $\triangleright$ 2 GEMMs vs. eigenbasis + 1 small GEMM
- 4: $\mathbf{num} \leftarrow \tilde{\mathbf{G}}^\top r$; $\mathbf{den} \leftarrow \text{rowsums}(\tilde{\mathbf{G}} \circ \tilde{\mathbf{G}})$
- 5: $t^2 \leftarrow \mathbf{num}^2 / \max(\mathbf{den}, \epsilon)$; $\mathbf{RSS} \leftarrow \max(\text{RSS}_0 - t^2, 0)$
- 6: $F \leftarrow t^2 / \mathbf{RSS} \cdot \nu$; $p \leftarrow F_{1,\nu}(F)$ $\triangleright$ batched survival function
- 7: **end for**
- 8: **return** $\{p, F, \text{RSS}, \hat{\beta}_g, \text{se}(\hat{\beta}_g)\}$

which equals v1's per-SNP regression exactly (Frobenius–Waugh–Lovell partialling out). Effect sizes and standard errors come in closed form,

$$\hat{\beta}_g = \frac{\tilde{g}^\top r}{\tilde{g}^\top \tilde{g}}, \quad \text{se}(\hat{\beta}_g) = \sqrt{\frac{\text{RSS}(g)}{(n - q - 1) \tilde{g}^\top \tilde{g}}}. \quad (8)$$

3.2 Batching

All per-SNP quantities are rows of matrix expressions. For a block $\mathbf{G} \in \mathbb{R}^{k \times n}$ of k SNPs:

$$\tilde{\mathbf{G}}^\top = (I - \mathbf{Q}\mathbf{Q}^\top) \mathbf{G}^\top, \quad \mathbf{num} = \tilde{\mathbf{G}}^\top r \in \mathbb{R}^k, \quad \mathbf{den} = \text{diag}(\tilde{\mathbf{G}}^\top \tilde{\mathbf{G}}) \in \mathbb{R}^k, \quad (9)$$

and (7) is applied elementwise. The transform $\mathbf{G}^\top$ is two GEMMs against the eigenbasis (one GEMM per side); the residualization is one small GEMM of cost $O(nqk)$. The scan is therefore $O(n^2m)$ dense linear algebra with BLAS-3 kernels, no Python-level per-SNP loop, and no per-SNP factorization; arithmetic defaults to `float32` (fitting stays `float64`).

3.3 Multi-df extensions

2-df genotypic tests (v1's ANOVA path) group SNPs by level count L and build $L-1$ indicator contrasts per SNP; the small $(L-1) \times (L-1)$ normal equations are solved *batched* over the block (stacked `solve`). GxE scans test the interaction column $g \circ E$ (1 df), or jointly $[g, g \circ E]$ (2 df) with the same batched two-column update. The stepwise MLMM rescans with the current cofactors appended to $\mathbf{X}$, selecting by $\text{BIC} = \ell - \frac{k}{2} \log n$ with backward elimination.

4 Large n : the kinship as a streaming operator

Following BOLT-LMM and LDAK-KVIK, above the dense regime the kinship is never materialized. `GenotypeKinship` wraps the genotype store and applies K only through streaming products,

$$KX = \frac{1}{m} \sum_{\text{blocks}} Z_b (Z_b^\top X), \quad (10)$$

with Z_b the Yang-2010 called-only standardized block; the `scale_k` normalization constants (mean off-diagonal, mean diagonal) are obtained from two cheap streaming passes and folded into the same products, so the operator equals the dense GRM to machine precision (verified to 10^{-14} in the

tests). Every large- n computation is operator-native: the randomized eigensolver, the SLQ solver of Section 6, the scan transform (through the truncated basis below) and gBLUP all consume mat-vec products only. Cost per product is $O(nm)$ tall-skinny GEMMs with $O(\text{block})$ working memory, replacing $O(n^2)$ storage and an $O(n^2m)$ materialization; the equivalence to the dense pipeline (variance components, likelihood, gBLUP, scan under gentle truncation) is tested directly.

Adaptive deflation. The SLQ solver deflates only *well-separated* extreme eigenvalues (Ritz values at least $4\times$ the bulk median): the analytic accounting for deflated directions is exact only for an invariant subspace, and subspace iteration cannot converge the boundary of a near-degenerate bulk. Spectra without outliers deflate nothing – the tight bulk is precisely the regime Gauss quadrature resolves well.

5 Large n : truncated randomized spectrum

Above $n \approx 8,000$ (or whenever the kinship is a streaming operator) the dense eigendecomposition is replaced by a randomized top- k approximation ($k = 2048$ by default). A Gaussian sketch $\Omega \in \mathbb{R}^{n \times (k+p)}$ ($p = 12$ oversampling) is powered by $\ell = 5$ subspace iterations, $\mathbf{Q}_0 \leftarrow \text{orth}(\mathbf{K}\Omega)$, $\mathbf{Q}_t \leftarrow \text{orth}(\mathbf{K}\mathbf{Q}_{t-1})$, and the Ritz extraction $B = \mathbf{Q}^\top \mathbf{K} \mathbf{Q}$, $B = W\Lambda W^\top$ yields $\hat{\Lambda}_k, \hat{U}_k = \mathbf{Q}W_k$ (Halko et al. 2011). The $V^{-1/2}$ map applied to any matrix A becomes

$$A = \delta^{-1/2}A + \hat{U}_k \left[\text{diag}((\hat{\lambda}_i + \delta)^{-1/2}) - \delta^{-1/2}I_k \right] \hat{U}_k^\top A \quad (11)$$

which is *exact* whenever the discarded eigenvalues vanish, costs two $n \times k$ GEMMs, and never forms an $n \times n$ matrix. The unexplained spectral mass $\tau = \text{tr}(\mathbf{K}) - \sum_{i \leq k} \hat{\lambda}_i$ is reported on the fit so the approximation can be audited; the same identity serves the scan, gBLUP and $\mathbf{V}^{-1}$ applications.

6 Large- n fitting: deflated stochastic Lanczos quadrature

Fitting δ without the full spectrum requires the spectral sums s_1, s_2, s_3, s_4 of Section 2. The SLQ solver estimates them from Gauss quadrature rules built by Lanczos iteration (Ubaru et al. 2017): with T_t the tridiagonal Lanczos matrix after t steps from start vector v ,

$$v^\top f(\mathcal{A}) v \approx \|v\|^2 \sum_{i=1}^t w_i f(\theta_i), \quad (\theta, w) = \text{eigenpairs of } T_t, \quad w_i = (S_{1i})^2, \quad (12)$$

exact for polynomials of degree $\leq 2t-1$ and spectrally accurate for analytic f such as $f_\delta(t) = (t + \delta)^{-1}$ and $\log(t + \delta)$. Trace terms use unbiased Rademacher probe averages ($\mathbb{E}[z^\top f(\mathcal{A})z] = \text{tr } f(\mathcal{A})$, probes pre-projected into the covariate complement for REML); the quadratic forms in $\mathbf{y}$ use one deterministic rule from the projected phenotype. All Lanczos runs use full reorthogonalization ($t \leq 96$ default, 12 probes).

Deflation. Gauss quadrature converges slowly when the spectrum mixes extreme outliers (population structure) with a tight bulk. Extreme eigenpairs of the operator are therefore estimated by a randomized top- d pass ($d = 128$) and handled analytically; the quadrature resolves only the deflated bulk. Because the deflated operator carries the removed directions as (near-)zero eigenvalues,

Algorithm 3 Deflated SLQ variance-component fit

Require: operator $\text{mv}(x) = \mathbf{S}\mathbf{K}\mathbf{S}x$; $\mathbf{y}$, $\mathbf{X}$; probes P , steps t , deflation d

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1:  $\hat{\lambda}, \hat{U} \leftarrow$  randomized top- $d$  of  $\text{mv}$ 
2:  $\text{mv}_d(x) \leftarrow \text{mv}(x) - \hat{U}(\hat{\lambda} \circ (\hat{U}^\top x))$ 
3:  $y_\perp \leftarrow \mathbf{S}\mathbf{y}$ ; decompose  $y_\perp = \hat{U}c + y_{\text{rest}}$ 
4: rules: one Lanczos( $\text{mv}_d, y_{\text{rest}}, t$ );  $P$  Lanczos( $\text{mv}_d, \mathbf{S}z, t$ )
5: for  $\delta$  on grid and within Brent bracket do
6:    $s_1(\delta) \leftarrow \sum_i c_i^2 / (\hat{\lambda}_i + \delta) + \mathcal{Q}_y[f_\delta]$ 
7:    $s_2(\delta) \leftarrow \sum_i \log(\hat{\lambda}_i + \delta) + \overline{\mathcal{Q}}_P[\log] - d \log \delta$  ▷ deflation correction
8:    $s_4(\delta) \leftarrow \sum_i (\hat{\lambda}_i + \delta)^{-1} + \overline{\mathcal{Q}}_P[f_\delta] - d/\delta$ 
9:   assemble  $\ell_R(\delta), \ell'_R(\delta)$  as in (??)
10: end for
11: return  $\hat{\delta}$  by bracketing + Brent;  $\hat{\sigma}_g^2 = s_1(\hat{\delta})/p$ 
```

the probe estimates must be corrected by their contribution $f_\delta(0)$:

$$\sum_{i>p-d} f(\lambda_i) = \left[\text{probe mean of } f_\delta\text{-rules} \right] - d f_\delta(0), \quad f_\delta(0) \in \{1/\delta, \log \delta, \delta^{-2}\}. \quad (13)$$

Quadrature nodes are floored at zero to guard against roundoff-negative deflated eigenvalues. On structure-rich kinships the resulting $\hat{\delta}$ agrees with the exact EMMA fit to a few percent (the REML profile is very flat near its optimum) and profile likelihoods agree to about one nat; the total cost is $O(n^2)$ per matvec times $O(\text{probes} \times \text{steps})$ matvecs, with no $O(n^3)$ factorization.

7 Kinships

Additive GRM. Per variant j , genotypes are standardized over *called* samples only, $z_{ij} = (g_{ij} - \bar{g}_j)/s_j$ for called g_{ij} and $z_{ij} = 0$ otherwise (Yang et al. 2010), and $\mathbf{K} = \sum_j w_j z_j z_j^\top / \sum_j w_j$ is accumulated over SNP blocks in `float64`. Optional weights w_j implement LDAK-style weighting; an optional SNP subset fits the null model from a marker subsample (LDAK-KVIK style) at reduced cost.

IBS. Identity-by-state similarity is accumulated as one-hot GEMMs per block: with H_a the indicator of genotype $a \in \{0, 1, 2\}$ over called entries, the shared-allele numerator is $\sum_a H_a H_a^\top$ and the denominator is $CC^\top$ with C the called-indicator, so $\mathbf{K}_{\text{IBS}} = S/C$ per pair.

Exact LOCO. Because the GRM is a sum of globally standardized outer products, the blocks accumulated for each chromosome $\mathbf{K}_c = m^{-1} \sum_{j \in c} z_j z_j^\top$ satisfy $\sum_c \mathbf{K}_c = \mathbf{K}$ *exactly* under a single global standardization, and the leave-one-chromosome-out matrix is the algebraic difference

$$\mathbf{K}_{\text{loco}}(c) = \frac{m \mathbf{K} - m_c \mathbf{K}_c}{m - m_c}, \quad (14)$$

rescaled to mean diagonal one. No per-chromosome restandardization is performed, hence no approximation. Windowed local/global pairs along the genome are the same construction restricted to a sliding interval.

8 Batched permutation thresholds

Permutation follows v1 semantics: the *raw* phenotype is permuted and variance components stay fixed at $\hat{\delta}$. Because permutation does not commute with the whitening transform ($P \neq P$), permuted phenotypes are transformed after permutation, $\mathbf{Y}_P = \mathbf{y}[\Pi]$, and residualized against the fixed $\mathbf{Q}$. All B permutations then flow through each SNP-block GEMM at once:

$$T = \tilde{\mathbf{G}}^\top R_P \in \mathbb{R}^{k \times B}, \quad F = \frac{T \circ T}{\text{den RSS}_{0,P}} \nu, \quad (15)$$

so a permutation costs a matrix multiply rather than a full scan, and the genome-wide 5% threshold is the 5th percentile of the per- permutation minimum p .

9 Two-kinship mixtures

For $\mathbf{V} = \sigma_{g_1}^2 \mathbf{K}_1 + \sigma_{g_2}^2 \mathbf{K}_2 + \sigma_e^2 I$, the ratio $\rho = \sigma_{g_1}^2 / \sigma_{g_2}^2$ is profiled on a grid of $\log \rho$ over $[-3, 3]$ (cascading refinements around the best bracket), fitting each candidate mixture $a\mathbf{K}_1 + (1-a)\mathbf{K}_2$ by the exact EMMA path of Section 2; variance shares are reported per matrix.

10 Missing data

Per-SNP sample subsetting (v1’s behavior) breaks block batching. The default policy is a per-variant missingness filter followed by mean imputation over called genotypes (EMMAX-consistent), fused into the int8-to-float block conversion (optionally a Numba kernel); `impute='none'` propagates NaN for analyses that prefer explicit handling.

11 Validation and measured performance

Correctness is certified against a float64 port of the v1 mathematics (`tests/_reference.py`): F statistics agree to $\text{rtol } 10^{-6}$, null p -values pass a Kolmogorov–Smirnov uniformity test, simulated causal loci are recovered with high power, the SLQ solver reproduces the exact fit (Section 6), and the bundled *A. thaliana* Atwell data runs end-to-end. Measured timings (development laptop, Apple silicon, 4 BLAS threads, AC power; archive `benchmarks/results/`):

Workload	Metric	Result
scan vs. v1-style per-SNP least squares ($n=2000$, $m=10k$)	speed-up	$123 \times (0.27 \text{ s vs } 33.7 \text{ s})$
scan, $n=500$, $m=100k$, float32	throughput	$2.5 \times 10^5 \text{ SNPs/s}$
scan, $n=2000$, $m=100k$, float32	throughput	$4.8 \times 10^4 \text{ SNPs/s}$ ($3.3 \times \text{float64}$)
500 permutations $\times$ 5k SNPs, $n=1500$ (batched)	throughput	$1.7 \times 10^6 \text{ perm-SNPs/s}$

12 References

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